The Goldbach Survivor-Gap Problem for Two-Cloud Primorial Masks

A Finite Jacobsthal-Type Reduction of Binary Goldbach

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Abstract

This paper isolates a single finite combinatorial problem naturally attached to binary Goldbach. For an even integer N and a sieve level z, set

$$Q(z) = \prod_{p \le z} p$$

and define the Goldbach survivor mask

$$\mathcal{G}_z(N) = \{x \bmod Q(z) : (x, Q(z)) = 1,\ (N-x, Q(z)) = 1\}.$$

Equivalently, for every prime $p \le z$, the residue x avoids the two-cloud forbidden set

$$F_p(N) = \{0,\ N \bmod p\}.$$

The Chinese remainder theorem gives the exact full-period count

$$|\mathcal{G}_z(N)| = \prod_{p \le z}(p - c_p(N)), \qquad c_p(N) = \begin{cases} 1, & p \mid N, \\ 2, & p \nmid N. \end{cases}$$

Thus the average spacing of survivors is of logarithmic-square scale, up to the usual Goldbach singular factor. However, the maximum gap cannot be of this scale: complete sieving removes the consecutive interval $[2, z]$, so the empty-run gap is at least $z - 1$. The correct finite target is therefore not a Cramer-type $(\log z)^2$ maximum-gap bound, but a Goldbach-scale subquadratic bound.

Let $g_z(N)$ denote the maximum length of a consecutive interval, modulo $Q(z)$, containing no element of $\mathcal{G}_z(N)$. The central problem formulated here is to prove, at the Goldbach complete-sieve scale $z \asymp \sqrt{N}$, that

$$g_z(N) = o(N),$$

or more concretely $g_z(N) = o(z^2)$. We prove the exact reduction: if this subquadratic survivor-gap statement holds for all sufficiently large even N, then strong Goldbach follows for all sufficiently large even N, with only finite checking and explicit small-prime boundary channels left outside the large-prime interior. The proof is purely finite: a survivor $x \in (z, N-z)$ at level $z \ge \sqrt{N}$ forces both x and $N - x$ to be prime.

The paper also records the survivor-polynomial formulation. In $\mathbb{Z}[X]/(X^{Q(z)} - 1)$, the mask polynomial

$$S_{z,N}(X) = \sum_{x \in \mathcal{G}_z(N)} X^x$$

factors into lifted local CRT masks. For every interval I, the localized survivor count is the constant coefficient

$$R_z(N; I) = [X^0]_{Q(z)} W_I(X) S_{z,N}(X^{-1}).$$

Thus the gap problem is exactly a uniform coefficient-positivity problem for interval polynomials.

The new material in this version formulates three concrete routes suggested by this finite reformulation. First, the autocorrelation of the two-cloud mask has an exact CRT product formula, leading to a variance problem for interval counts. Second, residual-prime distribution can be averaged over even parameters N in a dyadic range, giving a realistic validation and partial-theorem target. Third, survivor-polynomial truncations suggest a computational-theoretical search for rigid finite minorants. The paper also formulates the finite torus-minorant problem: a pointwise lower-bound divisor weight, strong enough on the complete-sieve interval, would imply binary Goldbach. These are finite combinatorial routes, not assumed analytic inputs.

The paper also records numerical evidence and a configurable computational workflow. In a high-range scan, every even $N \leq 10^{10}$ except the trivial boundary-only case $N = 4$ had a complete-sieve interior representation; the largest observed offset of the first interior coordinate above $\lfloor\sqrt{N}\rfloor$ was 3464. These computations are validation evidence for the formulation only. They are not used as proof.

This paper does not prove Goldbach. Its contribution is a precise finite reduction and a sharply formulated candidate bridge: a subquadratic gap theorem for Goldbach two-cloud primorial survivor masks.

Role in the series. This preprint is intended as the next specialized paper in the MSI survivor-envelope program. It takes the finite survivor, windowed-envelope, boundary-channel, and survivor-polynomial tools of the previous papers and specializes them to the binary Goldbach diagonal. The paper is a reduction and problem-formulation paper, not a proof of the binary Goldbach conjecture.

Keywords. Goldbach conjecture; finite sieve; survivor gap; Jacobsthal problem; primorial modulus; Chinese remainder theorem; two-cloud mask; survivor polynomial; coefficient positivity; additive number theory; computational number theory.

1 Introduction

The binary Goldbach conjecture asks whether every even integer $N > 2$ is a sum of two primes. In the symmetric affine notation used in the previous papers of this program, one sets

$$n = N - 2, \qquad H = \{0\},$$

and obtains the two-coordinate packet

$$\mathcal{L}_{n,\{0\}}(i) = \{i + 1,\, n - i + 1\}.$$

Writing $x = i + 1$, this is simply

$$x, \qquad N - x.$$

Thus the binary Goldbach problem is a localized diagonal nonemptiness problem: one must find an x in the bounded interval $1 < x < N - 1$ such that both coordinates are prime.

The finite survivor, windowed-envelope, boundary-channel, and survivor-polynomial machinery used here is inherited from the global survivor-envelope framework of [3]. The present paper specializes those tools to the binary Goldbach two-cloud mask and isolates the remaining gap problem.

Classical treatments of Goldbach-type problems, sieve methods, and the general distribution of primes provide important background and calibration [6, 5, 10, 11]. The present paper uses none of these analytic theories as proof input. Its statements below are finite congruence identities, conditional finite reductions, or explicitly labelled finite problems.

The complete-sieve viewpoint makes the finite skeleton transparent. If $z \geq \sqrt{N}$, then any composite integer $m \leq N$ has a prime divisor $p \leq z$. Hence any integer $m \in (1, N]$ surviving all primes $p \le z$ is prime, except that prime values $m \le z$ are removed by the complete sieve itself and must be recorded as boundary channels. Therefore binary Goldbach can be separated into two finite pieces:

large-prime interior survivors

and

small-prime boundary channels.

This separation is exact and uses no analytic prime-distribution input.

The question is then whether the large-prime interior contains a survivor. At level z, define

$$Q(z) = \prod_{p \le z} p$$

and the Goldbach survivor mask

$$\mathcal{G}_z(N) = \{x \bmod Q(z) : (x, Q(z)) = 1,\ (N - x, Q(z)) = 1\}.$$

The mask is finite and periodic modulo $Q(z)$. It is described locally by the forbidden residue set

$$F_p(N) = \{0, N \bmod p\} \subset \mathbb{Z}/p\mathbb{Z}.$$

For an even N, local admissibility is automatic: modulo 2 the two forbidden classes collide, while for odd p at most two classes are removed.

The central quantity in this paper is not the global size of $\mathcal{G}_z(N)$, which is an immediate CRT product, but the maximum empty interval in its support. Let $g_z(N)$ denote the largest number of consecutive residue classes modulo $Q(z)$ containing no survivor. If, at the complete-sieve scale $z \asymp \sqrt{N}$, one can prove

$$g_z(N) = o(N),$$

then the interval $(z, N-z)$ must contain a survivor for all sufficiently large even N. This survivor gives a Goldbach representation with both primes larger than z. Small-prime representations are handled separately by the explicit channels $x = q$ or $N - x = q$, $q \le z$.

This is the one-tool reduction proposed here:

$$\boxed{\text{prove a subquadratic gap bound for Goldbach two-cloud survivor masks.}}$$

The average spacing of $\mathcal{G}_z(N)$ is of logarithmic-square size, but a maximum gap of that size is impossible in the complete mask: the interval $[2, z]$ is always removed by the condition $(x, Q(z)) = 1$. The correct target is therefore not $g_z(N) \ll (\log z)^2$, but rather

$$g_z(N) = o(z^2)$$

at the Goldbach scale $N \asymp z^2$. This statement is much weaker than a Cramer-type maximum-gap bound, but still strong enough to imply the binary Goldbach conjecture outside a finite range.

The rest of the paper is organized as follows. Section 2 defines the two-cloud Goldbach mask and gives its exact CRT count. Section 3 defines the empty-run gap and explains the forced boundary lower bound. Section 4 proves the finite reduction from a subquadratic Goldbach-scale gap bound to binary Goldbach. Section 5 records the survivor-polynomial coefficient formulation. Section 6 places the problem beside the ordinary Jacobsthal problem. Section 7 formulates the CRT lift induction and the missing distribution lemma. Section 8 formulates the finite torus-minorant problem. Section 9 records three concrete routes suggested by the finite reformulation: autocorrelation and variance, averaged residual-prime distribution, and survivor-polynomial minorant search. Section 10 explains how ternary Goldbach serves as a known validation case for the additive-ladder picture. Section 12 lists the computational tasks suggested by the reduction.

2 The Goldbach two-cloud mask

Definition 2.1 (Primorial modulus)**.** For $z \geq 2$, let

$$Q(z) = \prod_{p \leq z} p.$$

All primes in this product are rational primes.

Definition 2.2 (Goldbach survivor mask)**.** Let N be an even integer. The level-z Goldbach survivor mask is

$$\mathcal{G}_z(N) = \{x \bmod Q(z) : (x, Q(z)) = 1,\ (N - x, Q(z)) = 1\}.$$

Equivalently,

$$x \in \mathcal{G}_z(N)$$

if and only if, for every prime $p \leq z$,

$$x \not\equiv 0 \pmod p, \qquad x \not\equiv N \pmod p.$$

For each prime $p \leq z$, define the local forbidden cloud

$$F_p(N) = \{0, N \bmod p\} \subset \mathbb{Z}/p\mathbb{Z}$$

and the local survivor set

$$G_p(N) = (\mathbb{Z}/p\mathbb{Z}) \setminus F_p(N).$$

Let

$$c_p(N) = |F_p(N)|.$$

Then

$$c_p(N) = \begin{cases} 1, & p \mid N, \\ 2, & p \nmid N. \end{cases}$$

In particular, for even N, the prime 2 contributes one forbidden class, not two.

Proposition 2.3 (Exact CRT count)**.** *For every even integer N and every $z \geq 2$,*

$$|\mathcal{G}_z(N)| = \prod_{p \leq z} (p - c_p(N)).$$

Equivalently,

$$\frac{|\mathcal{G}_z(N)|}{Q(z)} = \prod_{\substack{p \leq z \\ p \mid N}} \left(1 - \frac{1}{p}\right) \prod_{\substack{p \leq z \\ p \nmid N}} \left(1 - \frac{2}{p}\right).$$

Proof. The condition defining $\mathcal{G}_z(N)$ is a system of independent congruence avoidance conditions modulo the primes $p \leq z$. For each prime $p \leq z$, there are $p - c_p(N)$ admissible residue classes. The Chinese remainder theorem identifies a residue class modulo $Q(z)$ with a tuple of residue classes modulo the primes $p \leq z$. Multiplying the local counts gives the claim. □

Remark 2.4 (Average spacing). The product in Proposition 2.3 implies that the average spacing between survivors in a complete period is approximately of $(\log z)^2$ size, multiplied by the reciprocal of the usual Goldbach singular factor. This is only an average statement. The maximum empty interval can be much larger, and the complete sieve forces a boundary gap of length at least $z - 1$, as shown below.

At the Goldbach scale $z \asymp \sqrt{N}$, the number N cannot be divisible by many distinct primes $p \le z$, because

$$\prod_{\substack{p \le z \\ p \mid N}} p \le N \asymp z^2.$$

Thus the collision case $p \mid N$, where the two forbidden classes collapse to one, is sparse in the logarithmic sense

$$\sum_{\substack{p \le z \\ p \mid N}} \log p \le \log N.$$

This scale restriction is important. A uniform statement over arbitrary residues $N \bmod Q(z)$ would include degenerate cases such as $N \equiv 0 \bmod Q(z)$, where the mask reduces to the ordinary reduced residue system. The Goldbach problem only requires the diagonal regime $N \asymp z^2$.

3 The survivor-gap function

Definition 3.1 (Empty-run gap)**.** Let $S \subset \mathbb{Z}/Q\mathbb{Z}$ be nonempty. Define $g(S)$ to be the largest integer $L \ge 0$ for which there exists an interval of L consecutive residue classes modulo Q containing no element of S. For the Goldbach mask we write

$$g_z(N) = g(\mathcal{G}_z(N)).$$

Thus every interval of length $g_z(N) + 1$ modulo $Q(z)$ contains at least one Goldbach survivor.

This definition measures the longest empty run rather than the distance between successive survivors. The two conventions differ by one. The present empty-run convention is convenient because it translates directly into interval nonemptiness.

Proposition 3.2 (Forced boundary lower bound)**.** *For every even N and every $z \ge 2$,*

$$g_z(N) \ge z - 1.$$

Consequently, no uniform estimate of the form

$$g_z(N) \le C(\log z)^2$$

can hold for the complete Goldbach survivor mask.

Proof. For every integer $x \in \{2, 3, \ldots, z\}$, the number x has a prime divisor $p \le z$. Hence $(x, Q(z)) > 1$, so $x \notin \mathcal{G}_z(N)$. Therefore the interval $[2, z]$ contains no survivor and has length $z - 1$. Since $z/(\log z)^2 \to \infty$, a bound $C(\log z)^2$ is impossible. $\square$

Remark 3.3 (Boundary channels)*.* The lower bound in Proposition 3.2 is not a defect in the framework. It is the finite trace of the small-prime boundary-channel phenomenon. Complete sieving at level z removes prime values $q \le z$ because such values are divisible by themselves. Therefore a complete-sieve Goldbach proof must combine large-prime interior survivors with explicit small-prime boundary channels.

The decisive scale is therefore not the average spacing. At complete level $z \asymp \sqrt{N}$, the usable large-prime interior interval has length approximately

$$N - 2z \asymp z^2.$$

Thus the gap theorem needed for Goldbach is any estimate that is smaller than this length, for example

$$g_z(N) = o(z^2)$$

or a more explicit estimate such as

$$g_z(N) \ll z(\log z)^A.$$

The second form would be much stronger than needed, but it is compatible with the forced boundary gap $g_z(N) \geq z - 1$.

4 Finite reduction to binary Goldbach

Definition 4.1 (Interior and boundary at level z)**.** Let N be even and $z \geq 2$. The large-prime interior interval is

$$I_z(N) = \{x \in \mathbb{Z} : z < x < N - z\}.$$

The small-prime boundary channels are the possible representations

$$N = q + (N - q), \qquad q \leq z,$$

where q is prime and $N - q$ is prime.

Lemma 4.2 (Complete-sieve prime interior)**.** *Let N be even and let $z \geq \sqrt{N}$. If*

$$x \in I_z(N)$$

and

$$x \in \mathcal{G}_z(N),$$

then both x and $N - x$ are prime.

Proof. Since $x \in I_z(N)$, both coordinates satisfy

$$z < x < N - z,$$

so in particular

$$1 < x < N, \qquad 1 < N - x < N.$$

Suppose x were composite. Then $x \leq N$, so x has a prime divisor $p \leq \sqrt{x} \leq \sqrt{N} \leq z$. This contradicts $(x, Q(z)) = 1$. Hence x is prime. The same argument applies to $N - x$, using $(N - x, Q(z)) = 1$. ☐

Proposition 4.3 (Boundary-interior equivalence)**.** *Let N be an even integer and let $z \geq \sqrt{N}$. Then N has a Goldbach representation if and only if at least one of the following holds:*

(i) there exists $x \in I_z(N) \cap \mathcal{G}_z(N)$;

(ii) there exists a prime $q \leq z$ such that $N - q$ is prime.

Proof. If (i) holds, Lemma 4.2 gives a representation by two primes. If (ii) holds, the representation is explicit.

Conversely, suppose $N = a + b$ with a, b prime. If one of a, b is at most z, then (ii) holds. If both are greater than z, then take $x = a$. Both x and $N - x = b$ are greater than z, less than $N - z$, and coprime to $Q(z)$. Thus $x \in I_z(N) \cap \mathcal{G}_z(N)$, so (i) holds. ☐

Conjecture 4.4 (Goldbach-scale two-cloud gap conjecture)**.** *Let N range over even integers and set*

$$z_N = \lceil \sqrt{N} \rceil.$$

Then

$$g_{z_N}(N) = o(N)$$

as $N \to \infty$ through even integers. Equivalently, since $z_N^2 \asymp N$,

$$g_{z_N}(N) = o(z_N^2).$$

Theorem 4.5 (Conditional Goldbach implication)**.** *If Conjecture 4.4 holds, then every sufficiently large even integer is a sum of two primes. If, in addition, the remaining finite range is checked computationally, then binary Goldbach follows for all even $N > 2$.*

Proof. Let N be even and large, and set $z = z_N = \lceil \sqrt{N} \rceil$. The interior interval

$$I_z(N) = \{x \in \mathbb{Z} : z < x < N - z\}$$

has length

$$|I_z(N)| = N - 2z - 1 + O(1) = N + O(\sqrt{N}).$$

By Conjecture 4.4, for sufficiently large N,

$$g_z(N) < |I_z(N)|.$$

Since every interval of length $g_z(N) + 1$ modulo $Q(z)$ contains a survivor, $I_z(N)$ contains some $x \in \mathcal{G}_z(N)$. Lemma 4.2 then implies that both x and $N - x$ are prime. Thus N has a Goldbach representation. The finite remainder can be checked directly. □

Remark 4.6 (No analytic input)*.* The implication in Theorem 4.5 is purely finite. It uses only divisibility, the complete-sieve threshold $z \geq \sqrt{N}$, and an empty-interval bound for $\mathcal{G}_z(N)$. No Hardy–Littlewood, circle-method, Bombieri–Vinogradov, Elliott–Halberstam, or Bateman–Horn input is used.

5 Survivor polynomials and coefficient positivity

The gap problem can be expressed exactly as a coefficient-positivity problem in the cyclic group algebra.

Let

$$\mathcal{A}_z = \mathbb{Z}[X]/(X^{Q(z)} - 1).$$

For each prime $p \leq z$, let $e_p \in \mathbb{Z}/Q(z)\mathbb{Z}$ be the CRT idempotent satisfying

$$e_p \equiv 1 \pmod{p}, \qquad e_p \equiv 0 \pmod{q} \quad (q \leq z, q \neq p).$$

Define the local mask polynomial

$$M_{p,N}(X) = \sum_{a \in G_p(N)} X^{a e_p} \in \mathcal{A}_z.$$

Proposition 5.1 (CRT mask factorization)**.** *In $\mathcal{A}_z$, the Goldbach survivor polynomial*

$$S_{z,N}(X) = \sum_{x \in \mathcal{G}_z(N)} X^x$$

factors as

$$S_{z,N}(X) = \prod_{p \leq z} M_{p,N}(X).$$

Proof. Expanding the product gives monomials with exponents

$$\sum_{p \leq z} a_p e_p \pmod{Q(z)}, \qquad a_p \in G_p(N).$$

By the defining property of the idempotents, this exponent is the unique CRT class whose reduction modulo p equals a_p for every $p \leq z$. Thus the set of exponents appearing in the product is exactly $\mathcal{G}_z(N)$, each with multiplicity one. □

For an interval $I \subset \mathbb{Z}$, define the window polynomial

$$W_I(X) = \sum_{x \in I} X^x \in \mathcal{A}_z.$$

Let

$$R_z(N; I) = |I \cap \mathcal{G}_z(N)|.$$

Proposition 5.2 (Interval coefficient identity)**.** *For every finite interval I,*

$$R_z(N; I) = [X^0]_{Q(z)} W_I(X) S_{z,N}(X^{-1}),$$

where $[X^0]_{Q(z)}$ denotes the coefficient of the identity class in $\mathbb{Z}[X]/(X^{Q(z)} - 1)$.

Proof. Expanding gives

$$W_I(X) S_{z,N}(X^{-1}) = \sum_{x \in I} \sum_{r \in \mathcal{G}_z(N)} X^{x-r}.$$

The constant coefficient modulo $Q(z)$ counts the pairs (x, r) with

$$x \equiv r \pmod{Q(z)}.$$

This is exactly the number of integers $x \in I$ whose residue class modulo $Q(z)$ lies in $\mathcal{G}_z(N)$. $\square$

Corollary 5.3 (Gap as coefficient positivity)**.** *The inequality*

$$g_z(N) < L$$

is equivalent to the statement that for every interval I of length L,

$$[X^0]_{Q(z)} W_I(X) S_{z,N}(X^{-1}) > 0.$$

Thus the Goldbach-scale two-cloud gap conjecture is a uniform localized coefficient-positivity problem for the CRT-factored polynomial $S_{z,N}$.

6 Relation with the Jacobsthal problem

The ordinary Jacobsthal problem studies long intervals of integers all having a nontrivial common factor with a fixed modulus. For the primorial modulus $Q(z)$, this is the problem of bounding gaps in the reduced residue system

$$(\mathbb{Z}/Q(z)\mathbb{Z})^\times.$$

It is a one-cloud problem: for each prime $p \leq z$, only the class

$$x \equiv 0 \pmod{p}$$

is forbidden. Classical work of Iwaniec gives foundational bounds for this problem [9]; modern large-prime-gap constructions, such as [4, 12], also use covering congruences and sieving constructions that are conceptually adjacent to Jacobsthal-type questions.

The Goldbach mask is different. It is a two-cloud problem:

$$x \not\equiv 0 \pmod{p}, \qquad x \not\equiv N \pmod{p}.$$

The second cloud is not arbitrary; it is a translate of the first cloud by the same global parameter N. This correlation is the structural feature that one would have to exploit to beat a purely worst-case covering argument.

Problem 6.1 (Goldbach two-cloud Jacobsthal problem)**.** At the Goldbach scale $z = \lceil \sqrt{N} \rceil$, prove a subquadratic upper bound

$$g_z(N) = o(z^2)$$

for the two-cloud survivor mask

$$\mathcal{G}_z(N) = \{x \bmod Q(z) : x \not\equiv 0, N \pmod{p} \text{ for all } p \le z\}.$$

Remark 6.2 (Why the scale restriction matters)*.* Problem 6.1 is not the same as a uniform bound over all possible residues $N \bmod Q(z)$. Goldbach requires $N \asymp z^2$. This prevents N from being divisible by an arbitrary set of small primes whose product exceeds z^2. Therefore the true Goldbach problem is a restricted family of two-cloud masks, not the full worst-case family of all masks modulo $Q(z)$.

7 CRT lift induction and the missing lemma

The natural induction adds primes one at a time. Let

$$Q_y = \prod_{p \le y} p$$

and suppose $p > y$ is the next prime. Every residue class $a \bmod Q_y$ has p lifts modulo $Q_y p$:

$$a + tQ_y, \qquad t = 0, 1, \ldots, p-1.$$

Since Q_y is invertible modulo p, the congruence

$$a + tQ_y \equiv 0 \pmod{p}$$

removes exactly one value of t, and the congruence

$$a + tQ_y \equiv N \pmod{p}$$

removes exactly one value of t, unless the two values coincide. Hence each old survivor has at least $p - 2$ surviving lifts.

Lemma 7.1 (Lift deletion bound)**.** *Let $a \bmod Q_y$ survive the Goldbach mask at level y, and let $p > y$ be prime. Among the p lifts*

$$a + tQ_y \bmod Q_y p, \qquad 0 \le t < p,$$

at most two are removed by the new prime p. If $p \mid N$, at most one is removed.

Proof. The two new forbidden congruences are

$$a + tQ_y \equiv 0 \pmod{p}, \qquad a + tQ_y \equiv N \pmod{p}.$$

Because Q_y is invertible modulo p, each congruence has one solution in $t \bmod p$. The two solutions coincide exactly when $N \equiv 0 \pmod{p}$. □

Globally, Lemma 7.1 immediately recovers the product count. Locally, it is not enough. A short interval may see only a thin portion of the lift grid, and the two deleted lift levels may erase all visible lifts of some old residue classes. The missing lemma must control this local distribution.

Problem 7.2 (Residual-prime distribution lemma)**.** Let $y < z$, let I be an interval, and let

$$\mathcal{G}_y(N) = \{x \bmod Q(y) : (x, Q(y)) = (N - x, Q(y)) = 1\}.$$

For primes $p \in (y, z]$, prove upper bounds of the form

$$\#\{x \in I : x \bmod Q(y) \in \mathcal{G}_y(N),\ x \equiv 0 \text{ or } N \pmod p\}$$
$$\leq \left(\frac{2}{p} + \varepsilon_{y,p,I,N}\right) \#\{x \in I : x \bmod Q(y) \in \mathcal{G}_y(N)\},$$

with total error strong enough, after inclusion-exclusion or polynomial coefficient control, to imply

$$R_z(N; I) > 0$$

for every interval I of length $o(z^2)$ at the Goldbach scale.

A first-order union bound over the residual primes cannot be expected to close the problem in full generality, because

$$\sum_{y<p\leq z} \frac{2}{p}$$

can exceed 1. Any successful proof must therefore use more structure. The survivor-polynomial identity provides the exact inclusion-exclusion algebra, and the two-cloud relation $\{0, N\}$ is the finite correlation that must replace a purely arbitrary covering argument.

8 Torus minorants and the finite error barrier

We now isolate the finite positivity problem that remains after the Goldbach mask has been placed on the primorial torus. The purpose of this section is to separate completely proved finite identities from the additional minorant problem that would be needed to close the Goldbach application.

Let $d \mid Q(z)$ be squarefree. Define

$$\rho_d(N) = \prod_{p \mid d} c_p(N).$$

This is the number of residue classes modulo d for which

$$d \mid x(N - x).$$

Lemma 8.1 (Interval divisor-count lemma)**.** *Let $I \subset \mathbb{Z}$ be an interval. For every squarefree $d \mid Q(z)$,*

$$A_d(I, N) := \#\{x \in I : d \mid x(N - x)\} = \frac{\rho_d(N)}{d}|I| + O(\rho_d(N)).$$

The implied constant is absolute.

Proof. Modulo d, the condition $d \mid x(N - x)$ is the product of the local conditions

$$x \equiv 0 \pmod p \quad \text{or} \quad x \equiv N \pmod p \qquad (p \mid d).$$

By the Chinese remainder theorem, the number of such residue classes modulo d is

$$\rho_d(N) = \prod_{p \mid d} c_p(N).$$

Each residue class modulo d occurs in the interval I as $|I|/d + O(1)$ integers. Summing over the $\rho_d(N)$ residue classes gives the formula. □

Proposition 8.2 (Exact inclusion–exclusion)**.** *For every interval I,*

$$R_z(N;I) = \sum_{d|Q(z)} \mu(d)A_d(I,N).$$

Consequently, the formal main term is

$$|I|\prod_{p\le z}\left(1-\frac{c_p(N)}{p}\right).$$

Proof. The condition defining $R_z(N;I)$ is

$$(x(N-x),Q(z)) = 1.$$

Inclusion–exclusion over squarefree divisors of $Q(z)$ gives

$$\mathbf{1}_{(x(N-x),Q(z))=1} = \sum_{d|(x(N-x),Q(z))} \mu(d).$$

Summing over $x \in I$ gives the first identity. Substituting the main term from Lemma 8.1 gives

$$|I|\sum_{d|Q(z)} \mu(d)\frac{\rho_d(N)}{d} = |I|\prod_{p\le z}\left(1-\frac{c_p(N)}{p}\right).$$

□

The difficulty is not the main term. The difficulty is the accumulated error if one uses the full inclusion–exclusion expansion directly. A crude absolute bound gives

$$\sum_{d|Q(z)} \rho_d(N) = \prod_{p\le z}(1+c_p(N)),$$

which is much larger than the expected main term at the complete Goldbach scale. Thus the CRT identities alone do not prove positivity in the short interval

$$I_z(N) = (z, N-z), \qquad z = \lfloor\sqrt{N}\rfloor.$$

This is the finite error barrier.

8.1 A finite minorant problem

The required strengthening may be formulated as a finite minorant problem. We seek coefficients λ_d, supported on squarefree divisors $d \mid Q(z)$, such that

$$\sum_{d|x(N-x)} \lambda_d \le \mathbf{1}_{(x(N-x),Q(z))=1}$$

for every integer x, and such that the corresponding interval sum is positive:

$$\sum_{x\in I_z(N)}\sum_{d|x(N-x)} \lambda_d > 0.$$

Since the minorant is pointwise bounded above by the survivor indicator, positivity of this weighted sum implies

$$R_z(N; I_z(N)) > 0.$$

Using Lemma 8.1, the weighted sum is

$$\sum_{d|Q(z)} \lambda_d A_d(I_z(N), N) = |I_z(N)| \sum_{d|Q(z)} \lambda_d \frac{\rho_d(N)}{d} + O\left(\sum_{d|Q(z)} |\lambda_d| \rho_d(N)\right).$$

Thus a sufficient finite criterion is

$$|I_z(N)| \sum_{d|Q(z)} \lambda_d \frac{\rho_d(N)}{d} > C \sum_{d|Q(z)} |\lambda_d| \rho_d(N),$$

for an absolute constant C, together with the pointwise minorant condition.

Problem 8.3 (Rigid two-cloud minorant problem)**.** For every even $N > 4$, with $z = \lfloor \sqrt{N} \rfloor$, construct explicit coefficients $\lambda_d = \lambda_d(N, z)$, supported on squarefree $d \mid Q(z)$, such that

$$\sum_{d|x(N-x)} \lambda_d \leq \mathbf{1}_{(x(N-x),Q(z))=1}$$

for every integer x, and

$$|I_z(N)| \sum_{d|Q(z)} \lambda_d \frac{\rho_d(N)}{d} > C \sum_{d|Q(z)} |\lambda_d| \rho_d(N).$$

This is a finite combinatorial problem. It does not assume the Hardy–Littlewood conjecture, the Bateman–Horn conjecture, the Elliott–Halberstam conjecture, or any asymptotic prime-distribution estimate.

Proposition 8.4 (Finite minorant implication)**.** *If the rigid two-cloud minorant problem has a solution for an even integer $N > 4$, then N has a Goldbach representation.*

Proof. The minorant hypotheses imply

$$0 < \sum_{x \in I_z(N)} \sum_{d|x(N-x)} \lambda_d \leq \sum_{x \in I_z(N)} \mathbf{1}_{(x(N-x),Q(z))=1} = R_z(N; I_z(N)).$$

Therefore there exists $x \in I_z(N)$ such that

$$(x, Q(z)) = 1, \qquad (N - x, Q(z)) = 1.$$

Since $x > z$ and $N - x > z$, Lemma 4.2 implies that both x and $N - x$ are prime. Hence $N = x + (N - x)$ is a Goldbach representation. □

8.2 The two-level tradeoff

The existing full-period torus distribution can be localized only when the base period is not too large compared with the interval. Choose $y < z$ and let

$$Q(y) = \prod_{p \leq y} p.$$

If $Q(y) \leq |I_z(N)| \asymp z^2$, then the level-$y$ torus gives meaningful interval control. Since $Q(y) \approx e^y$, this requires roughly

$$y \lesssim 2 \log z.$$

But then the residual primes $y < p \leq z$ have total first-order deletion mass approximately

$$\sum_{y<p\leq z} \frac{2}{p} \sim 2 \log \frac{\log z}{\log y},$$

which eventually exceeds 1. Conversely, choosing y large enough to make this residual sum small makes $Q(y)$ much larger than the Goldbach interval, so the current torus distribution theorem no longer localizes.

This tradeoff explains why the present theory does not yet prove the needed gap bound. The missing ingredient is not another full-period CRT count, but a localized non-covering or minorant theorem exploiting the rigid relation

$$F_p(N) = \{0, N \bmod p\}.$$

8.3 Relation with sieve weights

The minorant formulation resembles Selberg-type lower-bound sieve constructions and, more distantly, the weighted architectures used in modern prime-gap work [5, 12]. In the present paper this analogy is only methodological. No standard sieve theorem is invoked to prove Problem 8.3. The point is instead to identify the exact finite inequality that a successful Goldbach-specific minorant would have to satisfy.

Remark 8.5 (Interpretation of the conditional implication). The conditional implication above should not be read as making the Goldbach problem disappear. It relocates the difficulty into a finite complete-sieve gap or coefficient-positivity statement. At the scale

$$z = \lfloor \sqrt{N} \rfloor, \qquad I_z(N) = (z, N - z),$$

the assertion

$$I_z(N) \cap \mathcal{G}_z(N) \neq \varnothing$$

is precisely the large-prime-interior Goldbach assertion. Thus a proof of a subquadratic bound for the relevant maximum gap, or of a pointwise minorant that forces this intersection, would be a proof of Goldbach in this finite language. The value of the reformulation is that it makes the obstruction explicit and suggests deterministic finite objects on which to work.

9 Three finite routes suggested by the reformulation

The complete-sieve gap formulation is essentially equivalent in difficulty to the large-prime-interior Goldbach problem. Nevertheless, it exposes finite structures that are less visible in the classical formulation. We record three routes that arise naturally from the survivor-polynomial and torus language. They should be understood as attack routes on the remaining obstruction, not as proofs of Goldbach.

9.1 Autocorrelation and variance of the two-cloud mask

Let

$$f_{z,N}(x) = \mathbf{1}_{\mathcal{G}_z(N)}(x), \qquad \mu_{z,N} = \frac{|\mathcal{G}_z(N)|}{Q(z)}.$$

For an integer lag h, define the full-period autocorrelation

$$C_{z,N}(h) = \frac{1}{Q(z)} \sum_{x \bmod Q(z)} f_{z,N}(x) f_{z,N}(x+h).$$

The CRT gives an exact product formula. For each prime $p \leq z$, let $\nu_p(h, N)$ be the number of distinct residue classes in

$$0, \quad N, \quad -h, \quad N - h \pmod{p}.$$

Then

$$C_{z,N}(h) = \prod_{p \le z} \frac{p - \nu_p(h,N)}{p}.$$

Indeed, the conditions $x \in \mathcal{G}_z(N)$ and $x + h \in \mathcal{G}_z(N)$ require x to avoid the four classes listed above modulo each prime p, with collisions counted only once.

For a generic lag h and an odd prime $p \nmid N$, the four classes are distinct, so the local two-point factor is

$$\frac{p-4}{p}.$$

The square of the one-point local density is

$$\left(\frac{p-2}{p}\right)^2.$$

The generic correlation ratio is therefore

$$\frac{(p-4)/p}{((p-2)/p)^2} = \frac{p(p-4)}{(p-2)^2} = 1 - \frac{4}{(p-2)^2} < 1.$$

Thus generic lags exhibit local anti-correlation. Survivors do not behave like independent random points at the local level; the two-cloud condition creates a repulsive two-point factor for most lags.

For a cyclic interval of length M, set

$$A_M(a) = \sum_{0 \le t < M} f_{z,N}(a+t).$$

A standard expansion gives the exact variance identity

$$\frac{1}{Q(z)} \sum_{a \bmod Q(z)} \bigl(A_M(a) - \mu_{z,N} M\bigr)^2 = \sum_{|h|<M} (M - |h|)\bigl(C_{z,N}(h) - \mu_{z,N}^2\bigr),$$

where the correlations are read modulo $Q(z)$. The main obstruction is the set of exceptional lags for which congruences such as

$$h \equiv 0, \ \pm N \pmod p$$

hold for many small primes. These lags create local collisions and can produce positive correlations. Controlling their total contribution strongly enough to pass from average interval behavior to a maximum-gap bound is the first concrete technical problem.

This route is the main direct attack on the complete-sieve gap problem: the autocorrelation formula identifies the obstruction quantitatively, while the variance identity gives a finite object that can be computed, estimated, and compared with the observed complete-window gap data.

9.2 Residual-prime distribution averaged over the parameter

A second route is to relax the pointwise requirement in N and average over even parameters in a dyadic range. Let N vary over even integers in

$$N \in [Z^2, 2Z^2],$$

with the complete level $z \asymp Z$. After sieving to an intermediate level $y < z$, one studies the base survivor set

$$\mathcal{G}_y(N) = \{x : (x, Q(y)) = 1, \ (N - x, Q(y)) = 1\}$$

inside the Goldbach window. The residual clouds are

$$x \equiv 0, N \pmod p, \qquad y < p \le z.$$

For fixed N, proving that these residual clouds do not cover all base survivors is the hard pointwise problem. When N varies, however, the classes $N \bmod p$ move through the residual primes, creating extra averaging not present in the fixed-parameter problem.

A representative target is a mean-square residual distribution estimate of the form

$$\sum_{\substack{N\in[Z^2,2Z^2]\\ N \text{ even}}} \left| \#\{x \in I_z(N) \cap \mathcal{G}_y(N) : x \equiv 0, N \pmod p\} - \frac{c_p(N)}{p} \#(I_z(N) \cap \mathcal{G}_y(N)) \right|^2$$

being small enough after summation over residual primes $y < p \le z$, and then over squarefree residual products. Such estimates would measure whether the residual clouds behave as predicted by the survivor-envelope model once the translate parameter N is averaged.

This route is a validation and partial-theorem target. Proving complete-sieve positivity for a positive-density or density-one set of even N would not prove Goldbach. It would, however, demonstrate that the finite torus reformulation produces distribution information at the Goldbach scale and would isolate the exceptional pointwise obstruction more sharply.

9.3 Rigid finite minorants from survivor-polynomial truncations

The third route is a computational-theoretical search for a pointwise certificate. Instead of importing a generic lower-bound sieve, one starts from the exact residual survivor-polynomial product. After a base level y, the residual factor has the schematic form

$$\prod_{y<p\le z} \big(I_p(X) - K_p(X)\big),$$

where $I_p(X)$ is the all-lift contribution and $K_p(X)$ records the forbidden local cloud at p. Expanding the product gives the exact residual inclusion–exclusion. The union-bound attrition estimate keeps only an absolute first-order loss; the polynomial product retains all higher intersections and alternating signs.

The proposed finite minorant search is as follows:

(i) choose N, z, y and build the exact local residual masks;

(ii) truncate or smooth the residual product by conductor, support size, residual degree, or squarefree level;

(iii) impose the pointwise minorant constraint

$$T(x) \le \mathbf{1}_{(x(N-x),Q(z))=1};$$

(iv) maximize the certified interval mass

$$\sum_{x\in I_z(N)} T(x);$$

(v) search for stable coefficient patterns and then attempt to prove those patterns symbolically.

This differs from a standard Selberg lower-bound sieve in one essential respect: the candidate weights are generated from the exact two-cloud survivor polynomial rather than chosen only from generic divisor-count considerations. The special structure

$$F_p(N) = \{0, N \bmod p\}$$

is retained throughout. The open question is whether this rigid translate structure permits a finite pointwise minorant unavailable in a generic dimension-two sieve. This route is the most direct path toward a pointwise Goldbach certificate, but it is also the route most exposed to the parity barrier.

9.4 Roles of the three routes

The three routes play different roles in the program. The autocorrelation route is the main technical attack on the complete-sieve gap problem: it gives an exact CRT formula for the two-point function and isolates the exceptional lags that could support long gaps. The averaged-over-N residual-prime route is a validation and first partial-theorem target: it weakens the pointwise problem while testing the predicted distribution at the Goldbach scale. The finite minorant route is a discovery mechanism for a possible pointwise proof: survivor-polynomial truncations can be optimized computationally and then, if stable patterns emerge, attacked symbolically.

Thus the proposed progression is

$$\begin{gathered}\text{autocorrelation identifies the obstruction,}\\ \text{averaging over } N \text{ tests the predicted distribution,}\\ \text{and survivor-polynomial minorants search for a pointwise certificate.}\end{gathered}$$

None of these routes is presently a proof of Goldbach. Together they specify where the remaining finite work must occur.

10 Weak Goldbach as a validation benchmark

The ternary Goldbach theorem, proved by Helfgott [7, 8], states that every odd integer greater than 5 is a sum of three primes. In the present paper this theorem is not used as an input to the binary problem. It is used only as a validation benchmark for the additive-ladder viewpoint.

Binary Goldbach is a one-dimensional affine slice

$$x_1 + x_2 = N$$

inside a two-dimensional survivor space. Ternary Goldbach is a two-dimensional affine slice

$$x_1 + x_2 + x_3 = N$$

inside a three-dimensional survivor space. At a finite sieve level z, the ternary survivor count has the form

$$R_z^{(3)}(N; I) = \#\{(x_1, x_2, x_3) \in I^3 : x_1 + x_2 + x_3 = N,\ (x_j, Q(z)) = 1 \text{ for } j = 1, 2, 3\}.$$

At complete level, large-prime interior survivors give prime triples, while coordinates $x_j \le z$ are again explicit boundary channels.

The higher-dimensional slice has more room. Therefore, in numerical and finite survivor-envelope experiments, the ternary case should exhibit more stable localized survivor mass than the binary diagonal. This expectation agrees with the historical and analytic fact that the ternary problem is now known, while the binary problem remains open. The validation role is modest but useful: it tests whether the finite survivor machinery reflects the additive-dimensional hierarchy

$$\text{unary primality} \longrightarrow \text{binary additive diagonals} \longrightarrow \text{higher-dimensional additive hyperplanes.}$$

11 Numerical validation and configurable computation

A numerical scan was run in the complete-sieve language described above. The first-interior part of the scan searched, for each even N, for the first prime $x > \lfloor\sqrt{N}\rfloor$ such that $N - x$ is also prime. This is exactly the large-prime interior condition from Lemma 4.2.

The high-range run used the dense backend and tested every even $N \leq 10^{10}$. Apart from the trivial boundary-only case $N = 4$, every tested even integer had a complete-sieve interior representation. The reported classification was as follows:

$$\begin{aligned} \text{even } N \text{ tested} &= 4,999,999,999, \\ \text{interior positive} &= 4,999,999,998, \\ \text{boundary-only} &= 1, \\ \text{no representation found} &= 0. \end{aligned}$$

The worst observed offset of the first interior coordinate above $\lfloor\sqrt{N}\rfloor$ was

$$3464,$$

occurring at

$$N = 9265570214, \qquad \lfloor\sqrt{N}\rfloor = 96257, \qquad x = 99721.$$

Thus the first interior pair in that record offset case was

$$99721 + 9265470493 = 9265570214.$$

The largest first interior coordinate observed in the scan occurred at

$$N = 9997348832, \qquad \lfloor\sqrt{N}\rfloor = 99986, \qquad x = 102673,$$

with offset 2687. The worst ratio $x/\lfloor\sqrt{N}\rfloor$ remains the small case $N = 68$, where the first interior pair is $31 + 37$.

A separate complete-window gap computation was performed exhaustively for every even $N \leq 250000$, and by sampling above that range. In the exhaustive range, the worst complete-window empty run was

$$1439$$

at

$$N = 137414, \qquad z = 370.$$

In the high-range sampled data, the worst sampled complete-window empty run was

$$6233$$

at

$$N = 972350002, \qquad z = 31182.$$

The ratio in this sampled case was approximately

$$6.41 \cdot 10^{-6}.$$

These computations do not prove the gap conjecture. They are included only as validation evidence for the finite formulation: the complete-sieve interior is not a rare numerical event, and the observed first-interior offset is very small relative to N in the tested range.

A configurable C++ scanner, `Goldbach Lab v03`, was prepared to make the numerical workflow reproducible and adjustable. It supports a dense odd-only prime-bitset backend for exhaustive scans within ordinary 64-bit ranges, a deterministic Miller–Rabin backend for sparse or very large tests, and separate modes for first-interior scans, exact complete-window gap scans, and sampled gap scans. The intended output files are record cases, failure cases, sampled gap records, and a summary report; full per-even-N output is deliberately avoided for large limits.

The exhaustive first-interior scan to 10^{10} was run with a command of the form

```
goldbach_lab_v03 --mode first-interior --maxN 10000000000 --backend dense.
```

The program is computational evidence infrastructure only. It is not part of the proof of any theorem in this paper.

12 Computational tasks

The reduction suggests a focused computational program for the MSI Survivor Lab. For each even N, set

$$z = \lceil \sqrt{N} \rceil.$$

Then compute or estimate:

(1) the local collision profile
$$c_p(N) = 1 + \mathbf{1}_{p \nmid N}, \qquad p \le z;$$

(2) the global count
$$|\mathcal{G}_z(N)| = \prod_{p \le z} (p - c_p(N));$$

(3) the first large-prime interior survivor in
$$(z, N - z);$$

(4) the small-prime boundary channels $q \le z$, $N - q$ prime;

(5) the largest observed empty run in restricted windows;

(6) the layered certificate obtained by taking a base modulus $Q(y) \le |I|$ and then applying residual-prime attrition or polynomial inclusion-exclusion;

(7) the ternary analogue as a validation comparison.

A useful command-line workflow would classify each N into one of the following categories:

interior survivor found,
boundary representation found,
layered certificate found,
unexplained by current certificate.

The last category does not mean that Goldbach fails. It means only that the current finite certificate did not prove positivity for that instance.

13 Conclusion

The finite survivor framework turns binary Goldbach into a sharply formulated support-spacing problem. The relevant object is the two-cloud primorial mask

$$\mathcal{G}_z(N) = \{x \bmod Q(z) : (x, Q(z)) = (N - x, Q(z)) = 1\}.$$

At complete level $z \geq \sqrt{N}$, any survivor in the interior interval $(z, N-z)$ gives a Goldbach representation by two primes. Therefore, a Goldbach-scale subquadratic empty-run bound

$$g_z(N) = o(z^2), \qquad z \asymp \sqrt{N},$$

would imply binary Goldbach outside a finite range.

The paper does not prove this gap bound. It identifies it as the central finite problem and records the exact algebraic forms available for attacking it: CRT mask factorization, interval coefficient positivity, complete-sieve boundary separation, lift-by-prime induction, finite torus minorants, autocorrelation variance identities, averaged residual-prime distribution problems, and survivor-polynomial truncation searches. The remaining task is to prove that the correlated two-cloud forbidden structure $\{0, N \bmod p\}$ cannot cover an interval of Goldbach-scale length after the complete-sieve restriction is imposed.

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